

Distributed operating system

A Distributed operating system extends the concept of resource management and user friendly interface for shared memory computers. A distributed operating system appears to its users as a centralized operating system for a single machine, but it runs on multiple independent computers.

- Issues in distributed operating systems

1. Lack of Global Knowledge

Communication delays are at the core of the problem

- Information may become false before it can be acted upon
- these create some fundamental problems:
 - no global clock -- scheduling based on fifo queue?
 - no global state -- what is the state of a task? What is a correct program?



Naming

named objects: computers, users, files, printers, services

- namespace must be large
- unique (or at least unambiguous) names are needed
- logical to physical mapping needed
- mapping must be changeable, expandable, reliable, fast

Scalability

How large is the system designed for?

- How does increasing number of hosts affect overhead?
- broadcasting primitives, directories stored at every computer -- these design options will not work for large systems.
- Compatibility
 - Binary level: same architecture (object code)
 - Execution level: same source code can be compiled and executed (source code).
 - Protocol level: only requires all system components to support a common set of protocols.



Process synchronization

test-and-set instruction won't work.

- Need all new synchronization mechanisms for distributed systems.
- Distributed Resource Management
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 - Data migration: data are brought to the location that needs them.
 - distributed filesystem (file migration)
 - distributed shared memory (page migration)
 - Computation migration: the computation migrates to another location.
 - remote procedure call: computation is done at the remote machine.
 - processes migration: processes are transferred to other processors.
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- Security
 - Authentication: guaranteeing that an entity is what it claims to be.
 - Authorization: deciding what privileges an entity has and making only those privileges available.
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- Structuring
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 - the monolithic kernel: one piece
 - the collective kernel structure: a collection of processes
 - object oriented: the services provided by the OS are implemented as a set of objects.
 - client-server: servers provide the services and clients use the services.

Communication Networks

WAN and LAN

- traditional operating systems implement the TCP/IP protocol stack: host to network layer, IP layer, transport layer, application layer.
- Most distributed operating systems are not concerned with the lower layer communication primitives.
- Communication Models

message passing

- remote procedure call (RPC)

Message Passing Primitives

Send (message, destination), Receive (source, buffer)

- buffered vs. unbuffered
- blocking vs. nonblocking
- reliable vs. unreliable
- synchronous vs. asynchronous

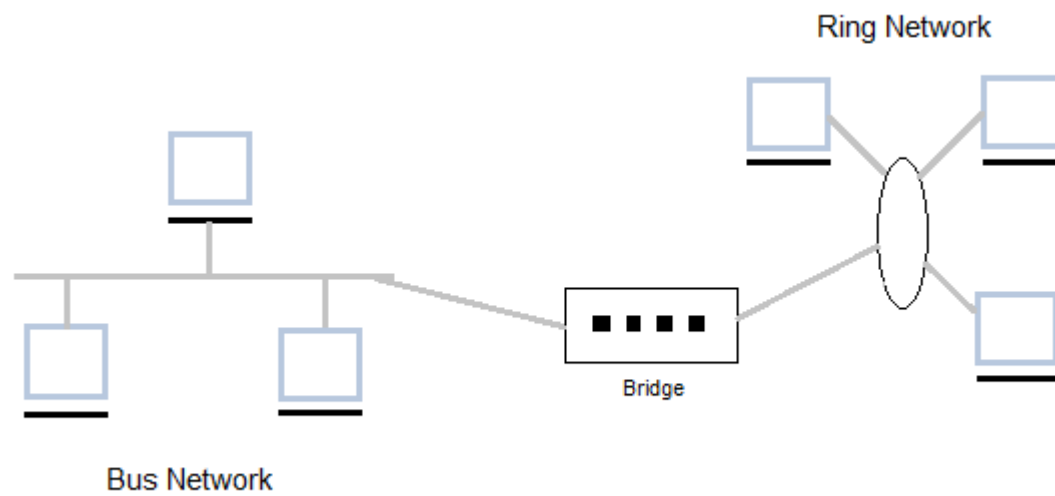
Communication networks

Local Area Network (LAN)

It is also called LAN and designed for small physical areas such as an office, group of buildings or a factory. LANs are used widely as it is easy to design and to troubleshoot. Personal computers and workstations are connected to each other through LANs. We can use different types of topologies through LAN, these are Star, Ring, Bus, Tree etc.

- LAN can be a simple network like connecting two computers, to share files and network among each other while it can also be as complex as interconnecting an entire building.
- LAN networks are also widely used to share resources like printers, shared hard-drive etc.





(Different Topologies interconnected in a Local Area Network)

Characteristics of LAN

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- LAN's are private networks, not subject to tariffs or other regulatory controls.
- LAN's operate at relatively high speed when compared to the typical WAN.
- There are different types of Media Access Control methods in a LAN, the prominent ones are Ethernet, Token ring.
- It connects computers in a single building, block or campus, i.e. they work in a restricted geographical area.

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- Applications of LAN

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- One of the computer in a network can become a server serving all the remaining computers called clients. Software can be stored on the server and it can be used by the remaining clients.
- Connecting Locally all the workstations in a building to let them communicate with each other locally without any internet access.
- Sharing common resources like printers etc are some common applications of LAN.

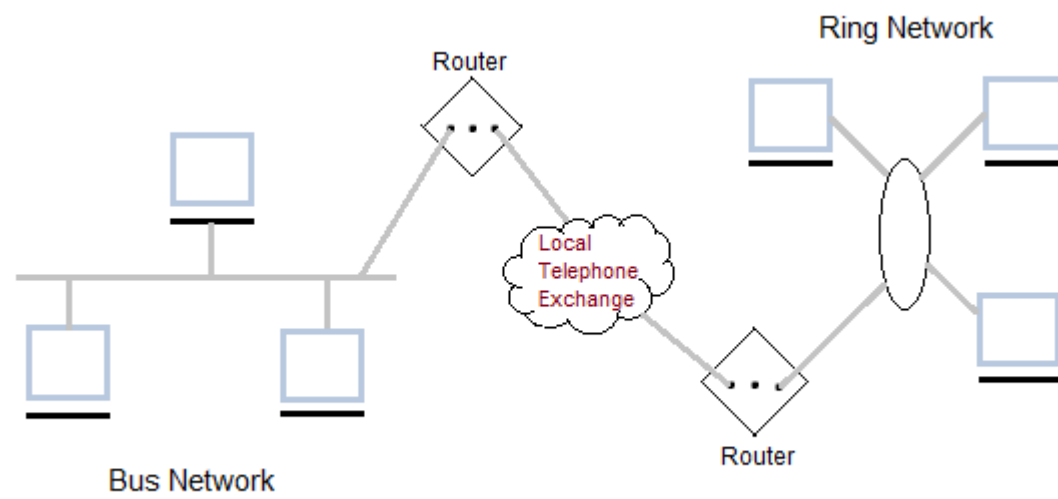
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Metropolitan Area Network (MAN)

It was developed in 1980s. It is basically a bigger version of LAN. It is also called MAN and uses the similar technology as LAN. It is designed to extend over the entire city. It can be means to connecting a number of LANs into a larger network or it can be a single cable. It is mainly hold and operated by single private company or a public company.

- Characteristics of MAN
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- It generally covers towns and cities (50 km)
- Communication medium used for MAN are optical fibers, cables etc.
- Data rates adequate for distributed computing applications.
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Wide Area Network (WAN)

It is also called WAN. WAN can be private or it can be public leased network. It is used for the network that covers large distance such as cover states of a country. It is not easy to design and maintain. Communication medium used by WAN are PSTN or Satellite links. WAN operates on low data rates.

Characteristics of WAN

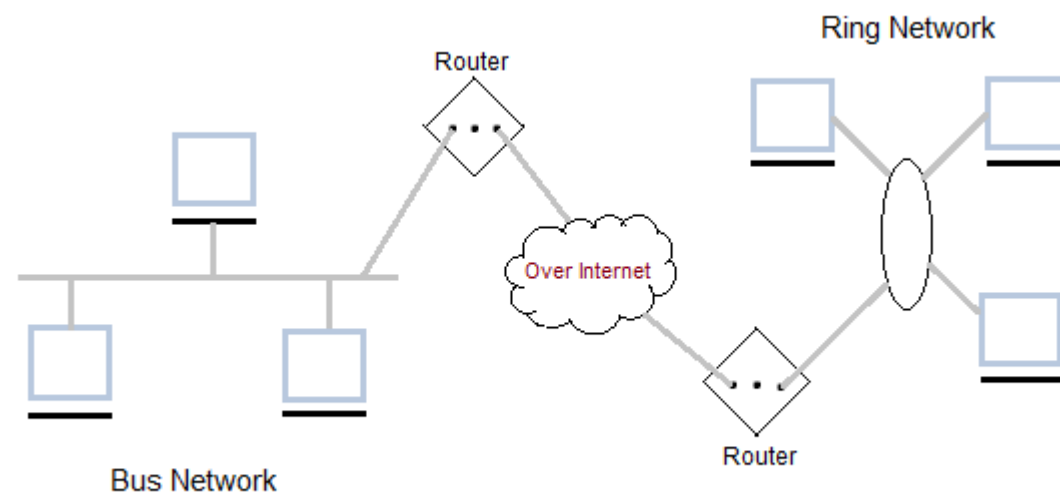
It generally covers large distances(states, countries, continents).

- Communication medium used are satellite, public telephone networks which are connected by routers.

Advantages of WAN

Covers a large geographical area so long distance business can connect on the one network.

- Shares software and resources with connecting workstations.
- Messages can be sent very quickly to anyone else on the network. These messages can have picture, sounds or data included with them(called attachments).
- Expensive things(such as printers or phone lines to the internet) can be shared by all the computers on the network without having to buy a different peripheral for each computer.
- Everyone on the network can use the same data. This avoids problems where some users may have older information than others.
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Communication primitives

The communication primitives are the high level constructs with which program use the underlying communication network. The communication primitives influence a programmers's choice of algorithms as well as the ultimate performance of the program. They influence both the case of a system and the efficiency of applications that are developed for the system.

- The message passing model

the message passing model provides two basic communication primitives. Namely, SEND and RECEIVE. The SEND primitives has two parameters. A message and its destination. The receive primitives has two parameters, the source of a message and a buffer for storing the message.

- Blocking and nonblocking primitives

In blocking messages are copied three times. From the user buffer to kernel buffer. From the kernel buffer on the sending

Computer to the kernel buffer on the receiving computer and finally from the buffer on the receiving computer to a user buffer.

In nonblocking primitives, the SEND primitives returns control to the user process as soon as the message is copied from the user buffer to the kernel buffer. The RECEIVE primitives signals its intention to receive a message and provides a buffer to copy the message. The receiving process may either periodically check for the arrival of a message or be signaled by the kernel upon arrival of message. The primary advantage of nonblocking primitives is that programs have maximum flexibility to perform computation and communication in any order they want.

- Synchronous primitives

With synchronous primitives a SEND primitive is blocked until a corresponding RECEIVE primitive is executed at the receiving computer.

- Asynchronous primitives

With asynchronous primitives the messages are buffered. A SEND primitive does not block even if there is no corresponding execution of RECEIVE primitives.

The corresponding Receive primitives can either be a blocking and non blocking primitives.

Remote procedure call

- While the message passing communication model provides a highly flexible communication ability programmers using such a model must handle the following details
- Pairing of responses with request messages.
- Data representation
- Knowing the address of the remote machine or the server
- Taking care of communication and system failures.

The handling of all these details in program makes the development of program for distributed computations difficult.



